SEMANTIC QUESTION-ANSWERING USING RETRIEVAL-AUGMENTED AI

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DECLARATION

We declare that the project work "SEMANTIC QUESTION ANSWERING USING RETRIEVAL-AUGUMENTED AI" was carried out by the following people:

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CERTIFICATION

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DEDICATION

We dedicate this project to God Almighty, whose grace, wisdom, and strength has guided us through to its completion.

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ABSTRACT

This study designs and implements a semantic question-answering system using retrieval-augmented generation (RAG) to address two critical limitations of large language models in educational contexts: hallucination of unverifiable responses and the inability to process lengthy documents within finite context windows. The system integrates document processing, vector embedding, semantic retrieval, and language model generation within a three-tier web application. User-uploaded PDF documents are chunked, embedded, and indexed in a vector store, enabling semantically relevant passages to be retrieved at query time and used to ground model responses, with source citations surfaced alongside every answer. By combining selective semantic retrieval with dynamic document management and transparent source attribution, the system produces trustworthy, context-aware answers drawn directly from user-provided materials, without requiring model fine-tuning or full-document processing.

Keywords: Retrieval-Augmented Generation, Semantic Question Answering, Large Language Models, Embeddings, Context Window

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CHAPTER ONE: INTRODUCTION

1.1 Background Study

Question Answering (QA) systems have steadily advanced from simple keyword-based approaches to architectures capable of interpreting user intent and semantic meaning embedded in natural language. Early QA systems mostly depended on keyword matching methods like bag-of-words, which offered quick retrieval but struggled to capture the complexity of natural language (Zhang & Zhao, 2021). As natural language processing (NLP) matured, researchers began focusing on methods that move beyond surface forms towards deeper semantic understanding (Chen et al., 2021). This shift recognizes that effective QA requires not only finding text fragments containing similar words but also reasoning about concepts, relationships, and contexts.

A notable development in this direction is the use of deep learning for answer selection. (Anggrayni et al., 2022) explored how bi-directional LSTMs with attention mechanisms, combined with vector embeddings, can represent both questions and answers in a shared semantic space (Anggrayni et al., 2022). Their findings showed that semantic similarity models outperform earlier keyword-matching techniques by better aligning retrieved answers with the actual intent of user queries. Such methods demonstrated improvements on benchmark QA datasets, showing how semantic modeling narrows the gap between user needs and retrieved responses (Anggrayni et al., 2022). Research on Semantic Link Networks (SLNs) shows that connecting ideas based on meaning can make QA systems more flexible and intelligent. (Xu et al., 2020) found that these networks help systems answer a wider variety of questions and handle complex reasoning (Xu et al., 2020). However, they also noted a tradeoff: while richer networks improve understanding, they require more computational resources (Xu et al., 2020). Overall, SLNs highlight the value of linking concepts, not just words, in building more robust QA systems.

Another strand of research emphasizes multi-layer semantic representation learning. (Chen et al., 2021) investigated how semantic representations at multiple levels can be integrated into QA models (Chen et al., 2021). Their work showed that incorporating layered semantic features leads to more accurate answer selection, particularly for context-dependent queries (Chen et al., 2021). This approach illustrates how deeper semantic representation supports greater reliability in aligning answers with question intent, offering a foundation for applications that demand precision.

While these advances show clear progress in QA research, the educational domain presents unique challenges and opportunities. Learners often pose questions using colloquialisms, discipline-specific terminology, or partial formulations that differ from textbook phrasing (Xu et al., 2020). Traditional keyword-based retrieval, while fast, frequently produces voluminous or irrelevant results that do not address learners' needs (Xu et al., 2020). In educational contexts, the problem is not merely retrieving documents but retrieving semantically appropriate and educative answers. To meet this challenge, QA research has turned toward education-specific tasks such as question classification.

Despite these developments, current educational QA systems still face key limitations. Research by Soong et al. (2024) demonstrates that when generative models are used without retrieval grounding, they often produce irrelevant or misleading responses (Wu et al., 2025). Similar risks occur in education, where uncurated generative output may be factually incorrect, or inconsistent with disciplinary conventions (Wu et al., 2025). Kee et al. (2024) also emphasize how hallucinations in large language model (LLM)-based systems lower trust, which is particularly detrimental in educational settings where reliability is paramount (Wu et al., 2025). In addition, even when reliable educational materials are available, current LLMs cannot effectively process them in their entirety. This is due to their finite context windows, which limit how much text they can consider at once. When long documents like multi-chapter textbooks or extensive research papers exceed these limits, models lose earlier information or overlook key sections, reducing coherence and factual accuracy (Liu et al., 2023). These limitations point to the need for QA systems that not only interpret meaning but also ground answers in verifiable sources while managing long, information-dense inputs efficiently.

The educational field provides a ground for advancing QA technologies. Effective systems must accommodate diverse learners, adapt explanations to different levels of prior knowledge, and provide correct responses (Wu et al., 2025). They must also bridge student queries with trusted instructional content while avoiding the pitfalls of generic retrieval or hallucinated responses. An effective way to address these limitations is through Retrieval-Augmented Generation (RAG), which has emerged as a promising paradigm. Rather than improving LLMs through further training, RAG boosts their performance by combining them with information retrieval methods (Xu et al., 2020). Essentially, the approach retrieves relevant documents based on an input sequence and incorporates them as supplementary context for generating the output sequence (Xu et al., 2020). By combining the semantic retrieval of relevant content with generative models capable of producing fluent, context-sensitive answers, RAG integrates the precision of information retrieval with the flexibility of language generation (Xu et al., 2020). Recent surveys affirm that RAG systems not only enhance factual grounding but also help mitigate the context window constraint by dynamically retrieving only the most relevant text fragments rather than processing entire documents at once (Gupta et al., 2025). Studies in related domains confirm that retrieval grounding improves accuracy, contextual appropriateness, and trustworthiness of generated answers (Wu et al., 2025).

For education, this integration holds particular promise. Semantic retrieval ensures that answers are drawn from trusted materials, while generation enables explanations to be tailored, elaborated, or simplified depending on learner needs (Wu et al., 2025). Intelligent systems could therefore provide explanations that directly target student misconceptions, and

AI assistants could synthesize domain-specific knowledge to deliver contextually appropriate responses that empower learner autonomy (Suwarningsih, 2020).

Nonetheless, educational QA remains an open research challenge since existing systems often fail to provide semantically grounded explanations across disciplines (Wu et al., 2025). Lexical retrieval and unguided generation cannot guarantee accuracy, coherence, or relevance (Wu et al., 2025). Bridging this gap requires a pipeline that grounds generative responses in retrievable, structured sources, supports explainable reasoning, and efficiently handles the large scale of educational materials within the limited context windows of modern LLMs (Suwarningsih, 2020).

1.2 Problem Statement

Current large language models (LLMs) often generate responses without referencing reliable sources, relying solely on pre-trained data. This leads to issues of trust, accuracy, and relevance, as responses may not reflect up-to-date or user-specific information. Furthermore, while it might seem feasible to provide entire text from materials like textbooks, lecture notes, or research papers containing thousands of pages, directly to an LLM, this approach is limited by the model's context window, which restricts how much text can be processed at once. When input exceeds that window, the model forgets earlier content, resulting in loss of important details and increased hallucination, where it produces believable but incorrect information. This research therefore proposes a semantic question answering system using retrieval-augmented AI that mitigates these issues by grounding responses in user-provided materials and selectively retrieving only the most relevant sections before generation, ensuring factual accuracy and context-aware explanations for learning and research.

1.3 Aim and Objectives of the Study

Aim

The aim of this study is to design and implement a semantic question answering system using retrieval-augmented AI, in order to provide accurate, explainable, and context-relevant answers for user consumption.

Objectives

To achieve this aim, the study will:

1. Design and implement the core RAG architecture, integrating document processing, vector embedding generation, semantic search, and a generator component to create an efficient question-answering pipeline.
2. Develop the user-facing application with key functionalities for user authentication, dynamic document upload and management, and an interactive chat interface that displays answers with source citations.
3. Validate the system's effectiveness by achieving a minimum score of 80% on a Faithful Accuracy metric, which requires answers to be both factually correct and fully grounded in the retrieved source material.

1.4 Significance of the Study

The significance of this study lies in addressing the twin challenges of factual reliability and limited context capacity in large language models. Current AI models often generate responses without verifiable grounding in reliable sources, leading to issues of trust and accuracy. Additionally, while educational materials such as textbooks, lecture notes, and research papers may contain thousands of pages, these cannot be processed in their entirety due to the finite context windows of language models. This creates a practical barrier where complete documents exceed what models can consider at once, resulting in loss of information and reduced answer quality.

By integrating retrieval-based grounding, the system enables learners and educators to obtain verifiable, context-aware explanations drawn from their own materials. The RAG-based approach ensures that responses remain connected to user-provided sources while strategically managing context limitations through selective retrieval of only the most relevant sections. This makes the tool particularly valuable for independent study, as it delivers reliable responses that maintain direct relevance to user content without requiring the model to process entire documents. The research advances AI development by creating systems that solve real-world educational needs while working within the practical constraints of current language model architectures.

1.5 Scope and Limitations of the Study

Scope

This study focuses on the design and implementation of a question-answering system using retrieval-augmented AI. The system is limited to integrating retrieval and generation components that manage the constraints of large language models' context windows by selectively fetching relevant content instead of processing entire documents. The study does not involve training or fine-tuning large language models; instead, it leverages existing GPT-based models to evaluate how effectively retrieval augmentation improves contextual grounding, factual accuracy, and relevance within user-provided learning materials.

Limitations

This study is subject to several methodological and technical constraints that define its scope and applicability. The system relies on OpenAI's pre-trained GPT models without fine-tuning, meaning response quality depends entirely on the model's inherent capabilities and the relevance of retrieved context rather than domain-specific optimization. The effectiveness of answer generation is bounded by the retrieval mechanism's accuracy, as irrelevant or incomplete retrieval directly compromises response quality since the generative model cannot compensate for missing or misaligned context.

This study addresses context window limitations through selective retrieval rather than architectural modifications to the language model itself. Alternative approaches such as fine-tuning, prompt engineering techniques, or hierarchical context management strategies fall outside the research scope. The system processes only text-based documents in PDF formats, meaning educational materials containing diagrams, mathematical notation, tables, or images are either excluded or processed without their visual components, potentially losing critical information. While the system architecture supports future expansion to multimodal inputs, this capability is not explored in the current study.

The study operates within the constraints of OpenAI API rate limits, token quotas, and cloud hosting capabilities. These infrastructure dependencies limit the scale of experiments, the volume of concurrent users that can be tested, and the comprehensiveness of performance benchmarking. System assessment is conducted using text-based educational datasets within a controlled environment, meaning performance characteristics under high-concurrency scenarios or across multiple subject domains simultaneously remain unexamined.

1.6 Research Questions

1. How can a retrieval-augmented generation architecture be designed to effectively integrate semantic retrieval with language model generation for question answering in educational contexts?
2. To what extent does grounding language model responses in user-provided educational materials improve the accuracy, relevance, and trustworthiness of generated answers compared to ungrounded generation?
3. What are the key technical and architectural considerations for implementing a RAG-based system that addresses the context window limitations of large language models while maintaining response quality?

CHAPTER TWO: LITERATURE REVIEW

2.1 Introduction

This chapter reviews existing literature relevant to semantic question answering and retrieval-augmented generation (RAG). The purpose is to establish the foundation for the study by examining how question answering (QA) systems have evolved, the role of semantic methods in improving accuracy, and how retrieval-augmented techniques address the limitations of large language models (LLMs). The review also considers the application of QA in education, where curriculum alignment, explainability, and learner trust are critical.

The chapter is organized thematically. It begins with an overview of the historical development of QA systems, from early keyword-based methods to modern transformer-based architectures. It then explores semantic approaches that aim to capture meaning beyond surface word matching, followed by a detailed discussion of RAG as a promising paradigm for grounding generative models in reliable sources. Attention is also given to how QA has been applied in educational contexts, highlighting both opportunities and persistent challenges. Evaluation methods, including standard benchmarks and human-centered criteria, are also reviewed to understand how QA performance is measured. Finally, research gaps are identified to show where current work falls short and how this study seeks to contribute.

2.2 Evolution of Question Answering Systems

Context-setting Question Answering (QA) has evolved through a sequence of paradigm shifts, each driven by advances in representation, data, and evaluation methodologies. Early work in QA was grounded in information retrieval (IR) and lightweight lexical matching, where the objective was to locate snippets containing potential answers rather than understand the meaning of the question. This retrieval-centric nature laid the groundwork for assessing QA systems in practical settings, as reflected in established evaluation tracks that emphasized extracting precise answer spans from text rather than returning broad document lists (Voorhees, 2001). Over time, natural language processing (NLP) techniques began to contribute more directly to QA by enabling syntactic and basic semantic interpretation. The shift toward machine learning (ML) and deep learning (DL) introduced data-driven representations such as word embeddings and neural architectures that could learn to map questions to passages and infer correct answers from context (Kazemi et al., 2022). As the volume and diversity of QA data grew, neural models demonstrated superior performance, yet their success depended critically on data availability, generalization, and the handling of reasoning beyond lexical cues (Kazemi et al., 2022). More recently, transformer-based architectures have emerged as a dominant force in QA systems, offering rich contextual representations and flexible conditioning on both questions and documents, these models have driven substantial gains in reading comprehension while simultaneously raising concerns about issues like hallucination (Tran et al., 2023). The progression from keyword-based systems to ML/DL and then to transformer-based QA illustrates a broad trajectory: increasing emphasis on meaning, context, and end-to-end learning, with ongoing concerns about reliability and interpretability. Taken together, these developments suggest that while keyword and purely linguistic approaches provided a solid foundation for early QA, current goals demand semantic grounding, good reasoning, and explicit alignment between user intent and retrieved knowledge (Kazemi et al., 2022; Tran et al., 2023).

2.2.1 Keyword-Based and Information Retrieval Approaches

Keyword-based and information retrieval approaches in the initial era of QA systems predominantly relied on information retrieval techniques that treated questions as lexical patterns to be matched against text. The primary objective was to retrieve passages likely to contain the answer, from which a human or an automated post-processing step would extract the exact span of words. This retrieval-centric mindset was formalized and evaluated through community-driven benchmarks such as the Text REtrieval Conference (**TREC**) QA track, which emphasized retrieving small snippets of text that answer the question rather than returning long document lists (Voorhees, 2001). The emphasis on text snippets as the unit of answer was instrumental in shaping early QA evaluation and in highlighting the limitations of purely lexical matching, including difficulty with recognizing different word forms and limited ability to handle synonyms or rephrased expressions (Voorhees, 2001).

2.2.2 NLP-Driven QA

The NLP-driven stage of QA introduced more sophisticated linguistic processing to interpret questions and locate relevant information. Parsing, part-of-speech tagging, named entity recognition, and shallow semantic representations were deployed to bridge the gap between surface form and intended meaning, enabling QA systems to encode syntactic and some semantic cues about both questions and potential answer passages (Tran et al., 2023). Early attempts in this phase sought to interpret natural language more directly, moving beyond monolithic keyword matching toward structured representations that could support more nuanced matching and inference. This transition reflected a growing recognition that language understanding, at least to a moderate depth, could be achieved through statistical NLP techniques, with hand-crafted features and probabilistic models guiding the alignment between question structure and textual evidence (Tran et al., 2023). While NLP-driven QA yielded improvements over pure IR, the depth of meaning captured remained constrained by relatively shallow semantics, limited world knowledge, and the absence of robust reasoning capabilities. The resulting systems often performed well on limited datasets or well-defined question types but struggled with questions requiring inference across multiple sentences or documents (Tran et al., 2023). In this era, progress depended on advances in linguistic annotation, parsing accuracy, and feature engineering to shape the inputs to downstream QA modules, rather than on fully end-to-end learning of question-to-answer mappings (Tran et al., 2023). The literature indicates that NLP-driven QA brought a significant shift toward interpretable linguistic signals and structured inference, but also highlighted persistent gaps in understanding nuanced intent and grounding answers in broader knowledge contexts (Tran et al., 2023; Zhang et al., 2022). This phase set the stage for a data-driven intensification, where machine learning would begin to automate feature extraction and inference in QA, ultimately enabling more scalable and generalizable systems as annotated data became more abundant (Kazemi et al., 2022). Overall, NLP-driven QA marked a critical transition from shallow lexical matching to more language-aware processing, while still grappling with limitations in deep semantic understanding and robust reasoning in the absence of large, diverse training data (Tran et al., 2023; Zhang et al., 2022).

2.2.3 Machine Learning and Deep Learning in QA

The ML and DL era transformed QA by adopting distributed representations of words and sentences, enabling models to learn semantic relationships from large corpora. Word embeddings such as Word2Vec provided continuous vector representations that captured distributional semantics, enabling better similarity judgments and more flexible matching beyond exact lexical overlap (Kazemi et al., 2022). This shift toward continuous representations laid the groundwork for neural architectures to learn end-to-end mappings from questions to passages and to predict exact answer spans within text. Recurrent neural networks (RNNs), long short-term memory networks (LSTMs), and convolutional neural networks (CNNs) became popular choices for QA tasks, offering the ability to encode sequences and hierarchical features that static lexical models could not capture (Kazemi et al., 2022). These DL models achieved notable performance gains on reading comprehension benchmarks (Kazemi et al., 2022). The literature also emphasizes that DL-based QA systems require substantial annotated data to achieve robust generalization, a requirement that has driven efforts in data collection, synthetic data generation, and domain adaptation to mitigate data scarcity in low-resource settings (Kazemi et al., 2022). Empirical results in ML/DL QA demonstrated clear advantages in representing linguistic context, enabling more nuanced question understanding and more accurate extraction of evidence, while still highlighting struggles in robust, multi-hop reasoning and explainability (Kazemi et al., 2022; Zhang et al., 2022). The machine learning advancement marked a decisive evolution from pattern-based and linguistically engineered features toward data-driven, representation-rich QA systems, but it also exposed the necessity for grounding and interpretability in complex QA scenarios (Kazemi et al., 2022; Zhang et al., 2022). Across this period, ML/DL QA benefited from improved corpora, pretraining techniques, and transfer learning, enabling cross-domain QA and more flexible adaptation to new languages and topics (Kazemi et al., 2022; Tran et al., 2023). The advantages of neural architectures were clear like capturing nuanced semantics, handling variable-length input, and learning from context, but researchers increasingly recognized the need to combine learning with explicit reasoning and retrieval strategies to answer questions that require information spread across documents (Tran et al., 2023; Zhang et al., 2022). In this light, the ML/DL phase did not merely outperform prior approaches; it also revealed the limitations of purely data-driven reasoning and motivated integration with retrieval and structured reasoning components to support more reliable QA (Zhang et al., 2022).

2.2.4 Transformer-Based QA

The most recent wave in QA uses transformer-based architectures, which provide powerful contextual representations and scalable, end-to-end learning for question answering. Transformer models enable rich conditioning on the entire text context and the question, allowing for nuanced understanding of meaning, long-range dependencies, and implicit world knowledge encoded in pretraining data (Kazemi et al., 2022; Tran et al., 2023). In the literature, there is an explicit shift toward neural network models with strong performance in machine reading comprehension and open-domain QA, driven by the capacity of transformers to model complex linguistic patterns and to leverage large-scale unsupervised pretraining followed by task-specific fine-tuning (Tran et al., 2023). This transition has accompanied substantial empirical gains, including improved ability to identify relevant evidence across documents, disambiguate questions, and generate coherent, contextually grounded answer outputs (Tran et al., 2023). Nevertheless, transformer-based QA is not without challenges: hallucinations and unsupported answer generation can arise when the model lacks reliable grounding in the retrieved passages, highlighting the need for robust grounding, evidence verification, and retrieval-augmented generation approaches (Tran et al., 2023). Empirical work further shows that the quality of QA heavily depends on high-quality data and appropriate alignment between question intent and retrieved content, reinforcing the importance of domain-adaptive training and curated evaluation suites (Kazemi et al., 2022). In parallel, studies of low-resource and multilingual settings underscore the value of architectures that can transfer knowledge across languages and domains, while also illustrating the difficulties of maintaining performance when training data are scarce or unevenly distributed (Tran et al., 2023). The transformer era thus represents a paradigm shift toward end-to-end, context-aware QA systems, while also foregrounding essential research challenges in grounding, reliability, and interpretability, particularly in open-domain scenarios (Kazemi et al., 2022; Tran et al., 2023). Within this context, a recurring theme is the integration of retrieval with generation or answer extraction to improve grounding and reduce the chance for hallucinations in transformer-based QA models (Tran et al., 2023). The central narrative is that transformer-based QA delivers strong contextual understanding and adaptability but requires careful architectural design and evaluation to ensure reliability and accuracy to source content (Tran et al., 2023). The culmination of these trends is a QA landscape in which performance arises from pipelines that couple powerful contextual models with retrieval strategies, explicit reasoning modules, and human-grounded evaluation, rather than from isolated improvements in any single component (Voorhees, 2001; Zhang et al., 2022).

2.3 Semantic Approaches in QA

The progression from keyword-based retrieval to semantic understanding represents a fundamental shift in how QA systems interpret and match questions with relevant information. Rather than relying solely on lexical overlap between queries and documents, semantic approaches attempt to capture the underlying meaning and conceptual relationships within text. This transition was motivated by the recognition that effective question answering requires understanding not just what words appear in a question, but what the question actually means and how that meaning relates to potential answers in the knowledge base. The limitations of keyword matching like its inability to handle synonyms, paraphrases, and implicit semantic relationships, drove researchers toward representation learning methods that could encode meaning in continuous vector spaces. These semantic methods have proven particularly valuable in educational contexts, where learners may express the same conceptual question using different terminologies, and where understanding the intent behind a query is essential for providing appropriate explanatory responses.

2.3.1 Embeddings for Semantic Representations

The development of distributed word representations marked a major advancement in semantic QA, enabling systems to move beyond surface-level word matching toward meaning-based similarity. Word embeddings such as Word2Vec and GloVe introduced the concept of mapping words into continuous vector spaces where semantically similar terms occupy nearby positions, allowing QA systems to recognize that "car" and "automobile" or "large" and "big" convey related meanings despite differing lexically. These embeddings are learned from large text corpora through techniques that capture *distributional semantics* which is the principle that words appearing in similar contexts tend to have related meanings. In QA applications, word embeddings enable more flexible matching between question terms and answer candidates, improving retrieval precision when queries use different vocabulary than the source documents.

Building on static word embeddings, contextual embeddings introduced by transformer-based models like BERT and GPT represent a significant leap in semantic representation quality. Unlike Word2Vec or GloVe, which assign a single vector to each word regardless of context, contextual embeddings generate different representations for the same word depending on its surrounding text. This context sensitivity proves crucial for QA tasks where word meaning depends heavily on usage, for instance, distinguishing "bank" as a financial institution versus a river bank. Karpagam et al. (2020) demonstrated that bidirectional LSTM models combined with attention mechanisms over embedded representations significantly outperform keyword-based approaches on benchmark datasets, showing how semantic modeling narrows the gap between user intent and system responses. These contextual representations form the foundation for modern semantic QA systems, enabling them to understand questions and match them to relevant content based on meaning rather than mere word overlap.

2.3.2 Semantic Similarity Search

Once questions and answers are represented as embeddings, the next challenge is determining which answers are most relevant to a given question. Semantic similarity search addresses this by measuring how close question and answer representations are in the vector space, rather than simply counting matching keywords. In this approach, both questions and candidate answers are encoded as vectors, and the system computes similarity scores, typically using measures like cosine similarity or dot products to identify which answers are semantically closest to the question. This fundamentally differs from keyword-based retrieval, which only recognizes relevance when the same words appear in both question and answer.

The key advantage of semantic similarity search is its ability to match questions with relevant answers even when they use completely different wording. For example, if a student asks "What causes rain?" and a textbook passage explains "Precipitation occurs when water vapor condenses," a keyword system would miss this match entirely. However, a semantic similarity approach can recognize that these express related concepts because their embeddings are close together in the vector space. This capability proves particularly valuable in educational settings, where students often phrase questions informally or use different terminology than their course materials.

The literature confirms that measuring semantic similarity in vector space leads to more accurate answer retrieval than traditional methods. By computing how close question and answer embeddings are to each other, systems can rank potential answers based on conceptual relevance rather than surface-level word matching.

2.3.3 Multi-Layer Representations

Advanced semantic QA systems leverage multi-layer representations that capture meaning at different levels of linguistic abstraction, from syntactic structure to high-level semantics. Yuan and Chen (2020) investigated how integrating representations from multiple semantic layers like syntactic features, distributional embeddings, and task-specific encodings improves answer selection accuracy. Their work demonstrated that models incorporating layered semantic features achieve better performance on nuanced or context-dependent queries compared to single-level representations, particularly for questions requiring inference across multiple sentences or conceptual domains. This multi-layer approach reflects the understanding that effective semantic representation requires capturing both local linguistic patterns and global conceptual relationships, as different layers of abstraction provide complementary signals for matching questions to answers.

Modern transformer-based architectures, like OpenAI's GPT-4 models, exemplify this multi-layer paradigm through their deep neural architectures comprising dozens of transformer layers. Each layer in these models processes increasingly abstract representations: early layers capture lexical and syntactic patterns, middle layers encode semantic relationships and contextual dependencies, while deeper layers perform high-level reasoning and conceptual integration. This hierarchical processing enables models like GPT-4 to handle complex educational queries that demand understanding at multiple linguistic levels simultaneously. Yuan and Chen (2020) emphasized that incorporating diverse semantic features at multiple layers leads to more reliable answer selection, offering a foundation for applications that demand both accuracy and interpretability. However, the literature acknowledges a persistent tradeoff: deeper semantic modeling yields better contextual understanding and more accurate results, but requires more sophisticated architectures and greater computational resources.

Despite this cost, multi-layer representations have become essential for handling the full complexity of natural language meaning in contemporary QA systems, particularly in educational contexts where questions often require reasoning across conceptual domains and inferring connections between disparate pieces of information.

While semantic models improve contextual matching and enable more flexible retrieval based on meaning rather than keywords, they still do not inherently ensure factual grounding or prevent the generation of plausible but unsupported answers. Even sophisticated embedding-based systems can retrieve semantically similar but factually incorrect content, and generative models can produce fluent responses that lack grounding in authoritative sources. This limitation has led researchers toward retrieval-augmented generation methods that explicitly combine semantic retrieval with constrained generation using an existing large language model, ensuring that system outputs are not only semantically coherent but also verifiably grounded in retrieved evidence.

2.4 Retrieval-Augmented Generation

The limitations of purely generative language models particularly their tendency to produce information but unverified responses have motivated the development of retrieval-augmented generation (RAG) as a hybrid approach that addresses the hallucination problem by grounding generation in retrieved evidence. RAG represents a paradigm shift in how language models access and utilize external knowledge, combining the precision of information retrieval with the fluency and adaptability of neural language generation. This section reviews the conceptual foundations of RAG, examines specific implementations across various domains, and analyzes how these systems have been applied to educational contexts.

2.4.1 Conceptual Foundations of RAG

Lewis et al. (2020) introduced the foundational RAG architecture that combines a neural retriever with a pre-trained sequence-to-sequence model, demonstrating that retrieval-augmented models achieve state-of-the-art results on knowledge-intensive NLP tasks while providing more factually accurate outputs than purely parametric approaches. Their work established the core RAG paradigm: given an input query, the system retrieves relevant documents from a knowledge base and conditions the generative model on both the query and retrieved context to produce grounded responses. This architecture addressed a fundamental limitation of large language models which is their reliance on static parametric knowledge that cannot be easily updated or verified by enabling dynamic access to external knowledge sources during inference.

The retrieval component in RAG systems typically employs dense retrieval methods using learned embeddings, where both queries and documents are encoded into a shared vector space that captures semantic similarity beyond lexical overlap. Karpukhin et al. (2020) demonstrated that dense passage retrieval using bi-encoders trained on question-answer pairs substantially outperforms traditional sparse retrieval methods like BM25 for open-domain question answering, achieving higher recall while maintaining computational efficiency through approximate nearest neighbor search. Their Dense Passage Retrieval (DPR) approach became foundational for many subsequent RAG implementations, showing that learned dense representations enable more accurate semantic matching between queries and relevant passages than keyword-based methods.

2.4.2 RAG Implementations Across Domains

The application of RAG has been explored across diverse domains, each revealing specific architectural considerations and implementation challenges. Asai et al. (2023) developed a self-RAG framework that enhances language model generation through retrieval and self-reflection, introducing mechanisms for the model to decide when retrieval is necessary and to critique its own outputs using reflection tokens. Their system demonstrated that teaching models to selectively retrieve and critically evaluate retrieved passages improves both factual accuracy and citation quality across diverse tasks, though at the cost of increased computational overhead and generation latency. This work highlighted a critical trade-off in RAG systems: while more sophisticated retrieval and reflection mechanisms improve output quality, they also increase system complexity and response time.

In the medical domain, Soong et al. (2024) investigated RAG for biomedical question answering, comparing GPT-3.5 and GPT-4 with and without retrieval augmentation using PubMed abstracts as the knowledge source. Their results showed that RAG substantially reduced factually incorrect responses from 46% error rates in ungrounded GPT-3.5 to 23% with retrieval augmentation while also improving the clinical relevance and specificity of generated answers. However, they identified persistent challenges: retrieval quality directly determined the upper bound on answer accuracy, and the system struggled when relevant information was distributed across multiple documents rather than contained in a single passage. These findings underscore the retrieval bottleneck in RAG systems, if relevant information is not retrieved, even sophisticated generators cannot produce correct responses.

Xiong et al. (2024) explored RAG for long-form question answering, developing systems that retrieve multiple passages and synthesize information across them to generate comprehensive answers. Their work revealed architectural challenges in managing multiple retrieved passages: the system must determine relevance, resolve contradictions, maintain coherence across diverse sources, and attribute specific claims to appropriate sources. They found that simply concatenating retrieved passages as context often led to incoherent generation, particularly when passages contained conflicting information or varied significantly in writing style. To address this, they developed iterative retrieval strategies where the model retrieves initial passages, generates a partial response, and then retrieves additional passages based on identified knowledge gaps which is a computationally expensive but more effective approach for complex questions requiring synthesis across multiple sources.

2.4.3 RAG in Educational Contexts

The application of RAG to educational question answering has received increasing attention, though implementations remain relatively limited compared to general-domain QA. Liu et al. (2024) specifically investigated textbook question answering using RAG, proposing a framework that combines dense retrieval over textbook chapters with GPT-based generation to answer student questions grounded in course materials. Their system indexed textbook content at the paragraph level, used a fine-tuned DPR model for retrieval, and instructed GPT-4 to generate answers strictly based on retrieved passages while citing specific textbook sections. Evaluation on a dataset of college-level biology questions showed that RAG improved answer accuracy by 34% compared to ungrounded GPT-4, with particularly strong gains on factual recall questions. However, the system performed poorly on questions requiring multi-step reasoning across chapters or integration of information from figures and diagrams, content types their text-only retrieval could not effectively handle.

Yan et al. (2024) surveyed applications of large language models in education more broadly, highlighting RAG as a promising approach for creating personalized learning assistants that ground explanations in curriculum-aligned materials. They noted that while LLMs demonstrate impressive general reasoning capabilities, their application in education requires mechanisms to ensure factual accuracy, curriculum alignment, and appropriate pedagogical framing which are requirements that RAG's grounding in authoritative sources helps address. However, they identified significant gaps in existing educational RAG systems: most operate over fixed textbook corpora rather than supporting user-uploaded materials, lack mechanisms for adapting explanation complexity to learner level, and provide limited explainability regarding how retrieved passages informed generated answers.

Swacha and Gracel (2025) conducted a comprehensive survey of RAG-based chatbots in education, categorizing systems by their knowledge sources, retrieval strategies, and pedagogical approaches. They found that most educational RAG implementations use relatively simple retrieval strategies like semantic similarity search over pre-chunked documents without sophisticated mechanisms for query expansion, retrieval ranking, or handling ambiguous questions. Common limitations included: difficulty with questions requiring synthesis across multiple documents, limited support for multimedia educational content, absence of curriculum-specific retrieval strategies, and lack of evaluation frameworks measuring pedagogical effectiveness beyond factual accuracy. They emphasized that educational RAG systems must balance competing objectives like factual correctness, pedagogical appropriateness, engagement, and accessibility, creating design challenges distinct from general-domain QA.

2.4.4 Technical and Architectural Considerations

Recent research has identified several technical challenges in RAG implementation that are particularly relevant to educational applications. Barnett et al. (2024) analyzed the retrieval bottleneck in RAG systems, demonstrating through controlled experiments that retrieval quality establishes a hard upper bound on generation quality, even state-of-the-art language models cannot overcome poor retrieval. They found that retrieval failures stem from multiple sources: semantic mismatch between query phrasing and document language, inadequate chunking strategies that split relevant information across passages, and insufficient contextual information in indexed passages to support accurate similarity matching. Their work highlighted the critical importance of chunking strategy in RAG systems: while smaller chunks enable more precise retrieval, they risk fragmenting contextual information; larger chunks preserve context but reduce retrieval precision and may exceed context window limits when multiple passages are retrieved.

The context window management challenge has been examined by several researchers. Liu et al. (2023) demonstrated that language models exhibit a "lost in the middle" phenomenon where they preferentially attend to information at the beginning and end of long contexts, often missing critical information in the middle even when it is relevant to the query. This finding has direct implications for RAG: simply concatenating multiple retrieved passages may not ensure the model effectively uses all retrieved information, particularly when many passages are retrieved or when the most relevant information appears in middle-ranked passages. They suggested architectural modifications such as re-ranking retrieved passages based on query relevance and strategically positioning the most relevant passages at the beginning or end of the context window.

Shuster et al. (2021) investigated knowledge grounding in dialogue systems, examining how language models balance retrieved information against their parametric knowledge. They found that models sometimes ignore or contradict retrieved passages when those passages conflict with the model's pre-trained knowledge, even when explicitly instructed to rely on retrieved context. This behavior poses risks in educational settings where curriculum-specific information may differ from general knowledge in the model's training data. Their experiments suggested that stronger grounding requires explicit architectural mechanisms such as constraining generation to derive from retrieved passages or using separate retrieval and generation scores to detect when models deviate from provided context.

2.4.5 Limitations of Current RAG Approaches

Despite substantial progress, current RAG implementations exhibit several limitations that motivate the present study. First, most systems operate over fixed, pre-indexed document collections rather than supporting dynamic document upload by end users. This limits their applicability to personalized learning scenarios where students work with their own textbooks, lecture notes, and materials. Second, chunking and indexing strategies remain largely domain-agnostic, treating educational materials like general documents rather than leveraging the inherent structure of textbooks, lecture slides, and academic papers.

Third, evaluation of RAG systems has focused primarily on factual accuracy metrics rather than pedagogical effectiveness criteria such as explanation clarity, conceptual correctness, and learner comprehension. Liu et al. (2024) noted that existing benchmarks measure whether systems produce factually correct answers but do not assess whether explanations are pedagogically appropriate for the target learner population. Finally, computational costs and latency remain practical barriers to deployment: RAG introduces overhead from retrieval operations, and serving multiple concurrent users requires significant infrastructure investment which are challenges that are particularly acute in educational contexts where systems must be accessible to individual learners without institutional resources.

These limitations reveal that while RAG offers a promising paradigm for grounding educational QA, substantial gaps remain in translating the approach into practical, user-accessible systems that support personalized learning with user-provided materials. The present study addresses these gaps by implementing a RAG-based system that enables document upload and evaluates both factual accuracy and answer grounding in educational contexts.

2.5 QA in the Educational Domain

The application of question answering systems in educational settings presents unique challenges that extend beyond technical accuracy. Educational QA must accommodate diverse learner backgrounds, align with curriculum objectives, provide pedagogically appropriate explanations, and build learner trust through reliable responses. Students often pose questions using informal language and varying levels of conceptual understanding, creating vocabulary gaps that systems must bridge while maintaining instructional appropriateness. Additionally, the scale of educational materials presents a fundamental technical challenge: comprehensive textbooks and lecture notes often contain hundreds or thousands of pages that exceed the context windows of large language models, making it impossible to process these materials in their entirety without loss of information (Liu et al., 2023).

2.5.1 Challenges in Educational QA

Educational question answering faces several interconnected challenges that distinguish it from general-domain QA. First, curriculum alignment requires that responses be both factually correct and pedagogically appropriate for the learner's level (Yan et al., 2024). Second, question diversity and ambiguity pose significant challenges as students often phrase questions informally or incompletely. Third, conceptual reasoning demands that systems go beyond surface-level keyword matching to understand underlying concepts and relationships (Chen et al., 2021). Fourth, personalization necessitates that explanations be tailored to individual learner needs (Suwarningsih, 2020). Finally, trust and reliability are paramount: inaccurate or misleading responses can reinforce misconceptions and undermine learning outcomes (Wu et al., 2025).

2.5.2 Existing Educational QA Systems

Various non-RAG approaches have been developed to address educational QA. Intelligent Tutoring Systems (ITS) use AI to provide personalized instruction through domain models (representing subject matter knowledge), student models (tracking learner progress), and pedagogical models (determining instructional strategies) (Suwarningsih, 2020). These systems have demonstrated effectiveness in K-12 and higher education, particularly for well-defined domains like mathematics and programming. However, ITS implementations often struggle with open-ended questions, require extensive domain engineering, and face challenges in scaling across diverse subject areas.

Traditional rule-based educational QA systems rely on carefully constructed knowledge bases and reasoning engines to answer student questions. While these approaches offer explainability and predictable behavior, they lack flexibility in handling natural language variation and require significant manual effort to develop and maintain domain-specific rules (Zhang & Zhao, 2021).

2.5.3 Limitations Motivating RAG Adoption

Current non-RAG educational QA approaches exhibit critical limitations that motivate the shift toward retrieval-augmented generation. First, lack of grounding in user-specific content represents a fundamental gap: most systems operate over fixed knowledge bases rather than the specific textbooks and materials that students actually use, creating misalignment with course terminology and frameworks (Yan et al., 2024). Second, insufficient adaptability persists as rule-based and traditional ITS systems cannot easily accommodate new content or respond flexibly to diverse question formulations (Zhang & Zhao, 2021). Third, hallucination and factual errors remain problematic in purely generative LLM-based systems without retrieval grounding (Wu et al., 2025). Finally, scalability and accessibility challenges limit deployment: systems requiring extensive setup, domain engineering, or computational resources may not be practical for individual learners (Suwarningsih, 2020).

These limitations motivate the adoption of RAG-based approaches that can ground responses in user-uploaded educational materials, provide verifiable source attribution, and leverage accessible cloud-based APIs to support diverse learners (Swacha & Gracel, 2025). The integration of retrieval mechanisms with generative models offers a pathway to address the unique requirements of educational QA while maintaining flexibility and accessibility.

2.6 Research Gaps Identified

The literature review reveals critical gaps in the implementation and evaluation of retrieval-augmented generation for educational question answering. The following three gaps motivate the present study's approach to developing a RAG-based system for user-provided documents.

1. **The Dynamic Personalization Gap:** Current educational RAG systems predominantly operate over fixed, pre-indexed textbook collections (Liu et al., 2024; Swacha & Gracel, 2025), limiting their application in personalized learning scenarios where students need to query their own specific materials, such as lecture notes or research papers. This study addresses this gap by implementing a system that supports dynamic user upload, indexing, and querying of personal document collections.
2. **The Retrieval Transparency Gap:** While retrieval quality is a known bottleneck (Barnett et al., 2024), existing educational RAG systems often function as "black boxes," providing generated answers without visibility into the retrieved evidence that informed them (Swacha & Gracel, 2025). This lack of transparency hinders user trust and the ability to diagnose errors. This study addresses this gap by implementing explicit source attribution, allowing users to verify the grounding of every AI-generated answer.
3. **The Pedagogical Faithfulness Gap:** Evaluations of educational QA systems focus heavily on factual accuracy but often neglect to measure the faithfulness of answers to the provided source material (Liu et al., 2024). A factually correct answer that is not grounded in the retrieved context is less valuable for curriculum-aligned learning. This study addresses this gap by explicitly evaluating the extent to which generated answers remain faithful to the user-uploaded documents, a key metric for reliability in educational settings.

2.7 Summary

This chapter reviewed literature on semantic question answering and retrieval-augmented generation. The review traced QA evolution from keyword-based retrieval to transformer-based architectures, showing increasing emphasis on semantic understanding alongside persistent challenges in grounding and reliability. Semantic approaches using embeddings and multi-layer representations enable meaning-based matching beyond keyword overlap. RAG emerged as a solution to hallucination by grounding generation in retrieved evidence, offering verifiable attribution and reduced errors, though facing retrieval bottlenecks and computational overhead.

Educational QA presents unique challenges including curriculum alignment, question diversity, conceptual reasoning demands, context window limitations when processing extensive materials, and trust requirements. Existing approaches like Intelligent Tutoring Systems and LLM-based assistants exhibit limitations such as lack of user-specific content grounding and hallucination risks. Two research gaps were identified: limited documentation of end-to-end RAG implementation for educational materials and insufficient understanding of how retrieval quality affects answer faithfulness in educational contexts.

The present study contributes to addressing these gaps by implementing a RAG-based QA system, evaluating answer grounding and faithfulness to source materials, and documenting implementation approaches and limitations. The next chapter details the methodology for designing, implementing, and evaluating this system.

CHAPTER THREE: METHODOLOGY

3.1 Introduction

This chapter outlines the methodology for developing a retrieval-augmented AI system. It covers the project's feasibility assessment, system requirements, architectural design, and the rationale for technology selection. The chapter provides a structured overview of how conceptual foundations were translated into an implementable system.

3.2 Feasibility Study

A feasibility study evaluated the system's practicality across technical, economic, and operational dimensions to ensure it could be developed and deployed within available resources and expertise.

3.2.1 Technical Feasibility

The project is technically feasible based on the following considerations:

1. **Available Technologies:** The system employs established tools such as NestJS, PostgreSQL, ChromaDB, and OpenAI models, all of which are stable, well-documented, and widely supported.
2. **Developer Expertise:** The required skills like TypeScript, RESTful API design, database management, and AI integration align with the developer's skills, supported by extensive documentation and learning resources.
3. **Infrastructure:** Cloud-based platforms like Vercel and Neon offer reliable, scalable, and cost-effective hosting for the application, database, and AI components.

3.2.2 Economic Feasibility

The system is economically viable due to minimal development and operational costs:

1. **Development Costs:** Core technologies (NestJS, PostgreSQL, ChromaDB) are open-source, eliminating licensing expenses. The primary investment is developer time.
2. **Operational Costs:** Costs remain low. The OpenAI API follows a usage-based model, while hosting on Vercel's free tier provides adequate resources for development and low-traffic deployment.

3.2.3 Operational Feasibility

The system is operationally practical and user-oriented:

1. **Accessibility:** The application runs on modern web browsers without requiring additional installations.
2. **Ease of Use:** A chat-based interface with document upload and simple authentication ensures an intuitive experience.
3. **Maintenance:** A modular architecture minimizes maintenance efforts.
4. **Deployment:** Cloud deployment enables effortless updates, scalability, and continuous availability.

3.3 System Requirements

This section outlines the system's functional and non-functional requirements, which define what the proposed retrieval-augmented AI platform must achieve and the standards it must meet in performance, security, and maintainability.

3.3.1 Functional Requirements

The system is expected to perform the following core functions:

1. Support user authentication through email-based magic links and Google OAuth 2.0.

2. Restrict access to protected resources for authenticated users only.
3. Manage user profiles containing basic information such as name and education level.
4. Enable users to upload, view, and delete PDF or DOCX documents within chat sessions.
5. Allow users to create, manage, and review their chat histories.
6. Process natural language queries and retrieve semantically relevant information using ChromaDB.
7. Generate AI-driven contextual responses grounded in retrieved document content via OpenAI's language model.
8. Persist conversation history, including user queries, AI responses, and document citations.
9. Deliver authentication emails using structured HTML templates for clarity and consistency.

3.3.2 Non-Functional Requirements

The system must adhere to the following performance, reliability, and design standards:

1. Maintain low latency and ensure timely responses under typical usage conditions.
2. Secure all communications over HTTPS and encrypt sensitive data such as tokens and credentials.
3. Validate inputs and implement error handling, logging, and recovery mechanisms to enhance reliability.
4. Support scalability through stateless service design and efficient resource management.
5. Maintain a modular TypeScript codebase with clear separation of concerns for ease of maintenance.

6. Provide automatically generated, up-to-date API documentation using OpenAPI/Swagger.

3.4 System Design

This section presents the design philosophy, overall architecture, and specific subsystem designs that showcases Lernen's technical foundation.

3.4.1 Design Philosophy and Principles

The proposed system's design would follow the Agile methodology and NestJS's architectural principles, emphasizing modularity and long-term scalability.

The system is guided by two core principles:

1. **Modularity:** The application is structured into independent modules, allowing features to evolve and scale without disrupting the overall system.
2. **Iterative Improvement:** Features are developed and refined in cycles, enabling continuous enhancement based on evolving requirements.

3.4.2 Architecture Overview

The proposed system would adopt a **three-tier architecture** consisting of the **presentation layer**, **application layer**, and **data layer**.

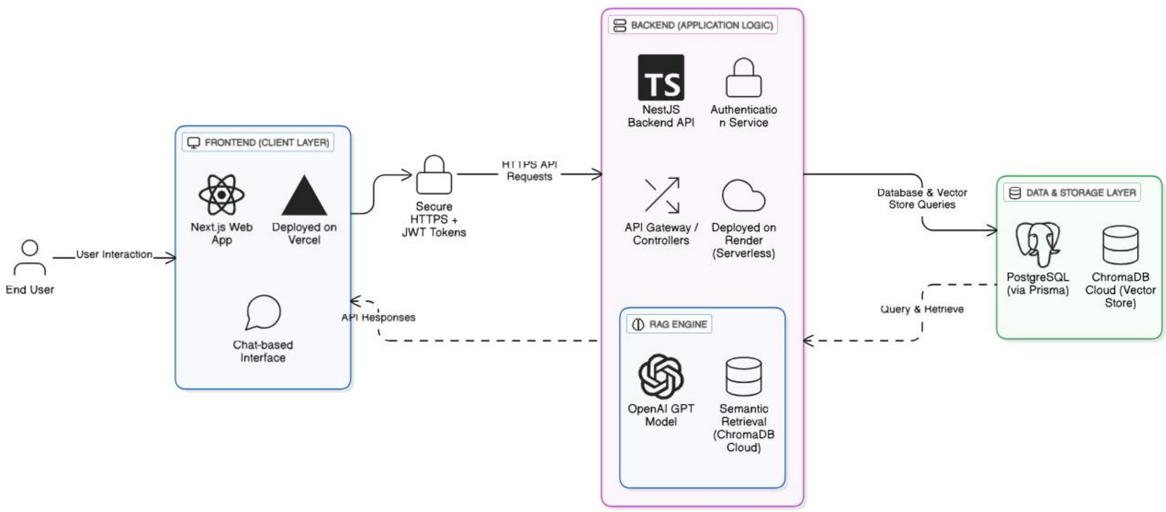

Figure 3.1 Diagram of the three-tier architecture overview

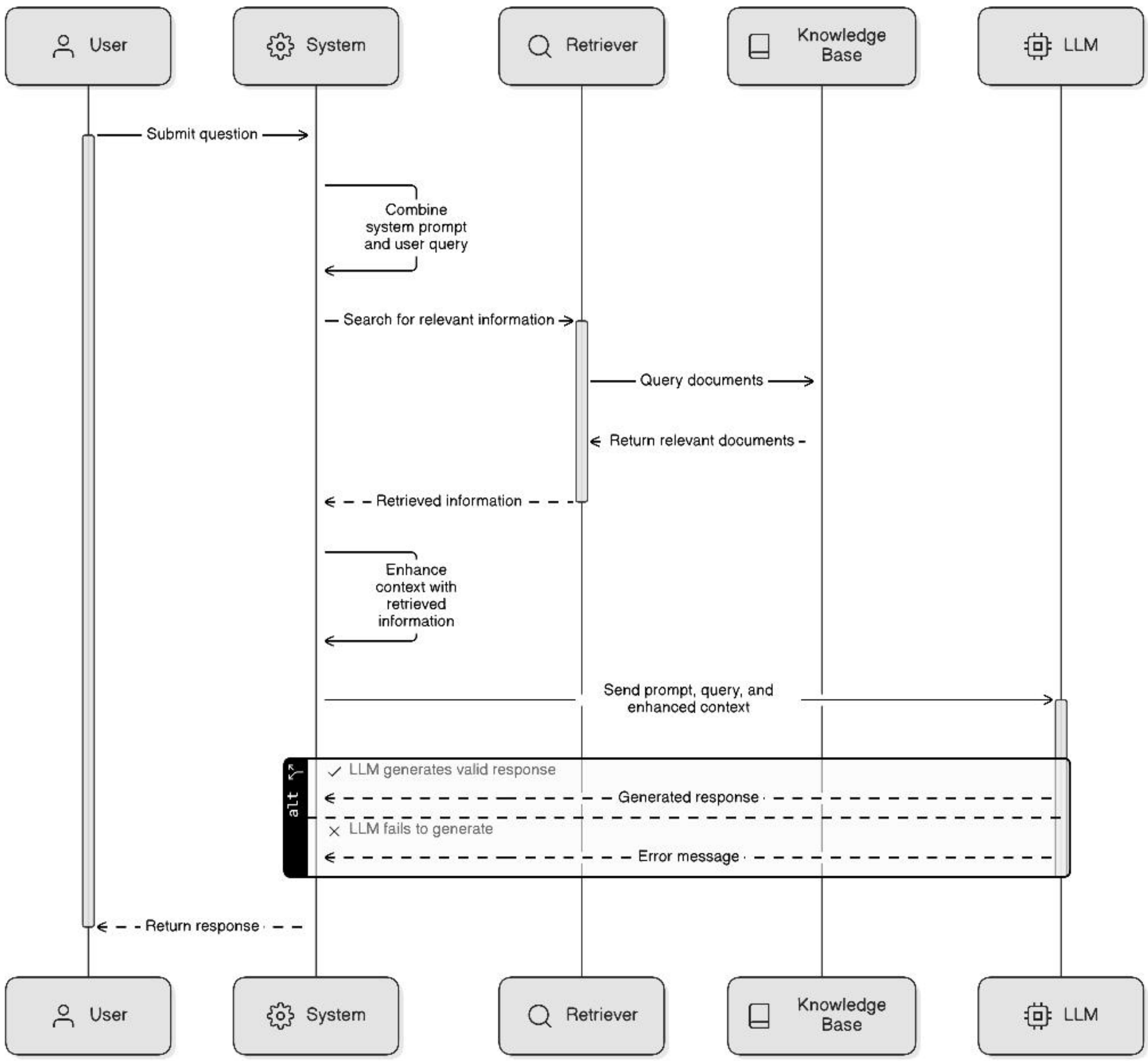

Figure 3.2 Sequence diagram of the RAG engine flow

3.4.3 Authentication Architecture

The proposed system would adopt a token-based authentication model designed for simplicity, scalability, and security. The system would support two authentication mechanisms: passwordless email sign-in using magic links and Google OAuth 2.0 for social login. Both methods issue JSON Web Tokens (JWTs) for maintaining stateless user sessions across client requests.

The authentication process separates identity verification from authorization. Email-based users authenticate through a time-limited magic link sent to their inbox, while Google users authenticate via OAuth consent. Upon successful verification, the server issues access and refresh tokens, which are used to securely access protected resources without storing session state on the server.

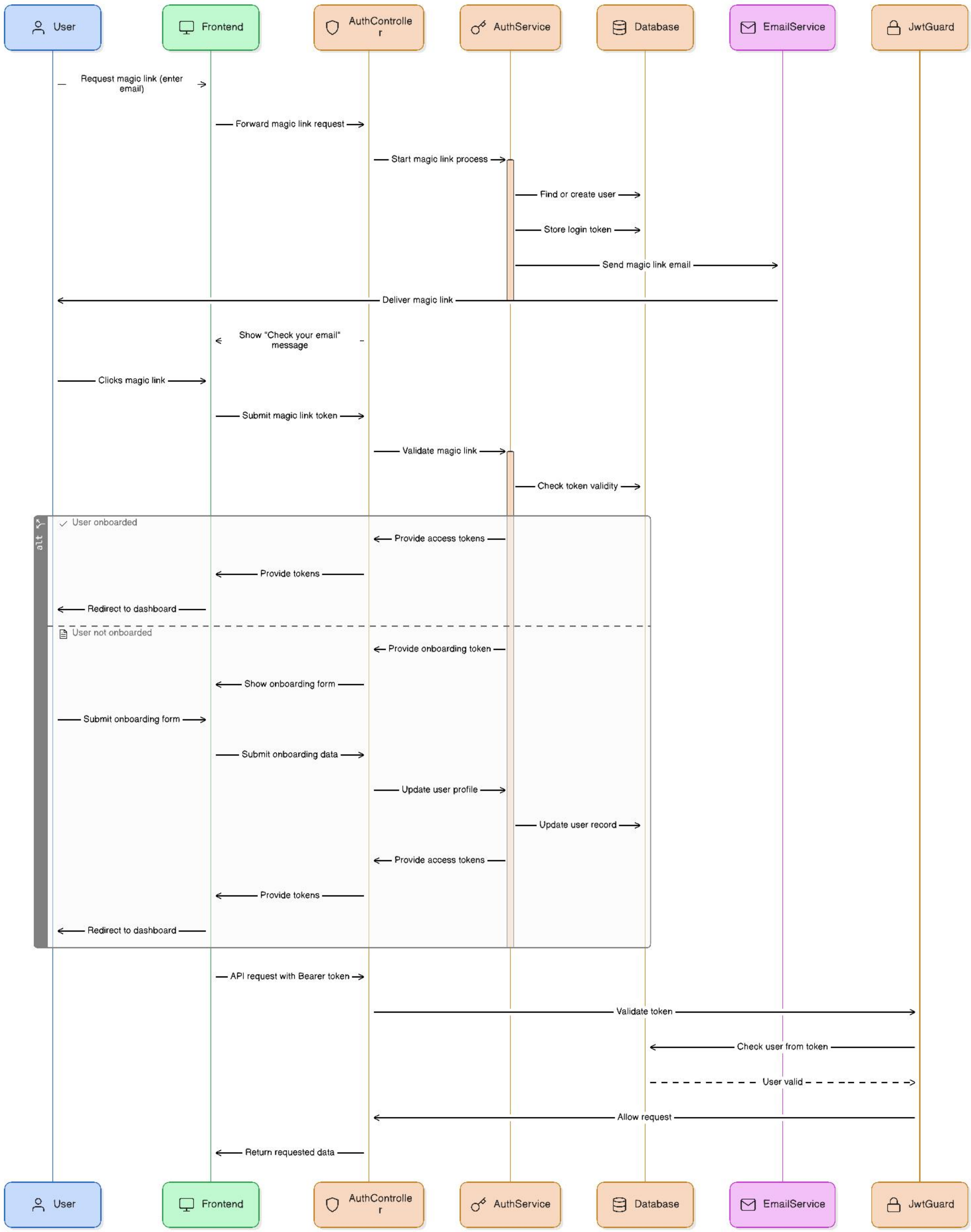

Figure 3.3 Sequence diagram of the magic link auth

This architecture enables seamless integration of multiple identity providers, reduces password management risks, and aligns with modern authentication standards suitable for distributed systems.

3.5 Technology Selection and Justification

The proposed system will employ modern, reliable technologies selected for their scalability, maintainability, and suitability for AI-powered learning environments:

1. **NestJS (Backend Framework):** Will be used to provide a modular and maintainable server architecture based on TypeScript.
2. **PostgreSQL with Drizzle ORM (Database):** Will manage structured data efficiently while supporting type-safe querying.
3. **ChromaDB (Vector Store):** Will enable semantic search and retrieval of user-uploaded documents.
4. **OpenAI Models:** Will provide intelligent, context-aware responses to user queries.
5. **JWT (Authentication):** Will ensure secure and stateless authentication through email magic links and Google OAuth.
6. **Nodemailer (Email Service):** Will handle passwordless login delivery through SMTP integration.
7. **LangChain and pdfjs-dist (Document Processing):** Will be adopted for parsing and chunking document text for semantic indexing.
8. **Security and Validation Tools:** A library like class-validator will be employed to ensure data integrity and protection.

CHAPTER FOUR: SYSTEM IMPLEMENTATION

4.1 Introduction

The implementation phase of this project involved translating the system design into a functional semantic question-answering platform. The system was developed using Next.js and React for the frontend, NestJS for the backend, ChromaDB for vector storage, and OpenAI's API for embedding generation and answer synthesis. User authentication was implemented through Passport.js supporting both email/password and Google OAuth, while the RAG pipeline enabled intelligent, context-grounded responses from user-uploaded documents.

This chapter details the implementation process, including the technologies and tools used, the core system architecture covering document processing and the RAG pipeline, and the user interface design for authentication, document management, and question-answering.

4.2 Core System Implementation

4.2.1 Document Processing and Embedding

The document processing pipeline handles PDF uploads, text extraction, chunking, and vector storage. When a user uploads a PDF file through the chat interface, the system first validates the file type and size (maximum 10MB). The PDF is then processed using pdfjs-serverless to extract text content page by page. Each page's text is extracted by combining all text items into a single string.

After extraction, the text is split into manageable chunks using LangChain's RecursiveCharacterTextSplitter with a chunk size of 1000 characters and an overlap of 200 characters to preserve context at chunk boundaries. Each chunk retains metadata including the page number and source document name.

The system generates 1536-dimensional vector embeddings for each chunk using OpenAI's text-embedding-3-small model. These embeddings are then stored in ChromaDB, a vector database optimized for similarity search. Each vector entry includes metadata such as document ID, filename, chat ID, user ID, page number, and source document name. The original PDF file is uploaded to S3 cloud storage asynchronously, and the document metadata is saved in the PostgreSQL database.

For retrieval efficiency, ChromaDB is configured with cosine similarity as the distance metric. The collection is cached in memory to avoid repeated initialization, and batch operations are used when adding embeddings to the vector store (maximum 300 embeddings per batch).

4.2.2 RAG Pipeline and Answer Generation

The RAG pipeline retrieves relevant context from uploaded documents and generates responses using a GPT model. When a user submits a question, the system first checks if there is conversation history beyond the recent messages stored in memory. If the conversation has more than four turns, the query is rewritten using GPT-4o-nano to incorporate relevant context from previous exchanges, making document retrieval more effective.

The rewritten query is converted into a 1536-dimensional embedding vector, which is used to search ChromaDB for the top 4 most similar document chunks. The search is filtered by chat ID, user ID, and selected document IDs to ensure only relevant documents are searched. The retrieved chunks are formatted with their source page numbers and filenames.

The system maintains conversation context through two mechanisms. Recent history stores the last four conversation turns in memory, while older conversations are compressed into summaries using GPT-4o-nano. Summaries are generated asynchronously every six turns to maintain context without overwhelming the model's context window. The most recent summary is retrieved from the database and included in the prompt.

The final prompt combines the user's education level, recent chat history, older chat summary, retrieved document context, and the current query. If the user is viewing a specific page in the PDF viewer, the page content is also included in the prompt for more contextually relevant responses. The prompt is sent to GPT-4.1-mini with a temperature of 1.0 for natural, varied responses.

The model's response is streamed back to the frontend using Server-Sent Events (SSE). The streaming process uses a two-step authentication flow: first, a session is created with authentication, storing the request parameters in Redis cache; then, the SSE stream is initiated without authentication, retrieving the session data from cache. This approach allows for authenticated streaming without exposing tokens in the EventSource connection. The response is displayed incrementally to the user as chunks arrive, and once complete, the full conversation turn (user message and assistant response) is saved to the PostgreSQL database.

4.3 User Interface Implementation

4.3.1 Authentication Interface (Sign-Up/Login)

The authentication interface supports two sign-in methods: email magic link and Google OAuth. For email authentication, users enter their email address and receive a temporary magic link that expires in 10 minutes. When clicked, the link verifies the token and authenticates the user. For Google OAuth, users are redirected to Google's authorization page, and upon successful authentication, a callback exchanges the authorization code for user information.

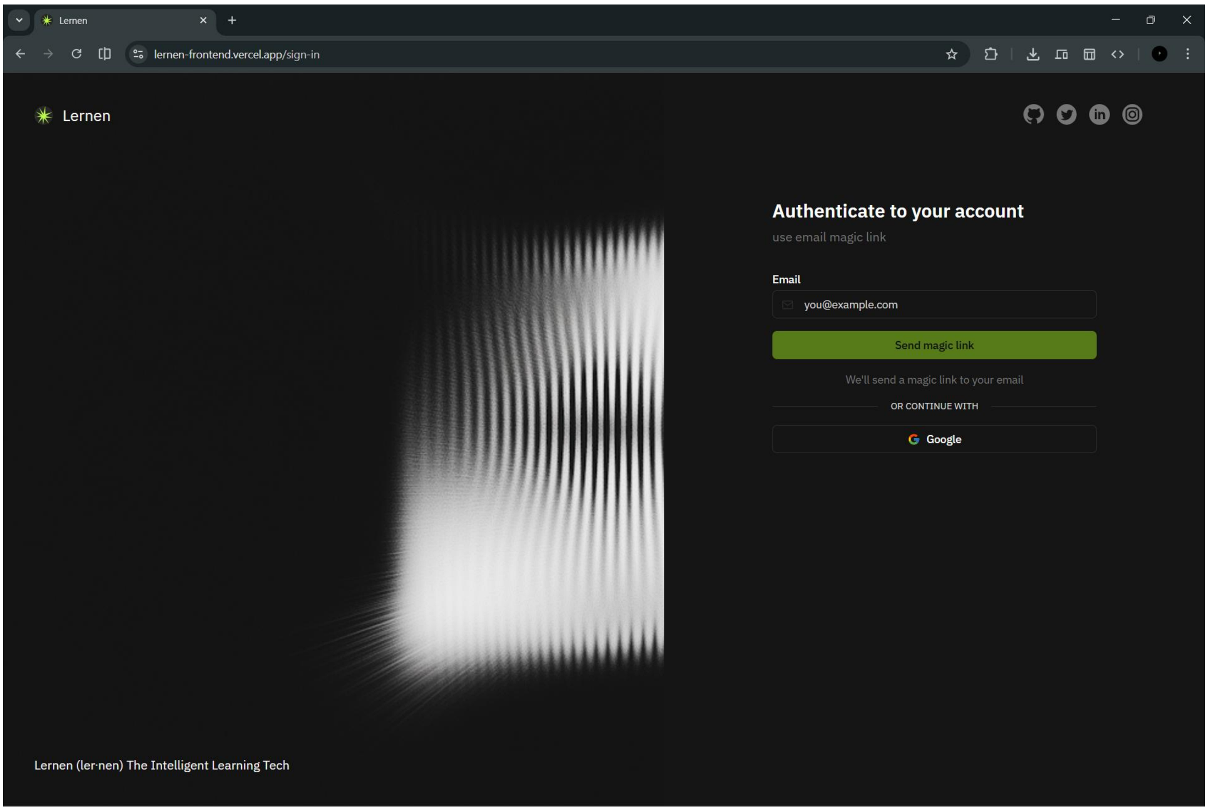

Figure 4.1 Sign up/Login page UI

The authentication system uses JWT tokens for session management, with access tokens stored in HTTP-only cookies for security. Refresh tokens are also stored securely and used to generate new access tokens when the current one expires. New users are redirected to an onboarding flow where they provide their name, education level, and learning preferences before accessing the main application.

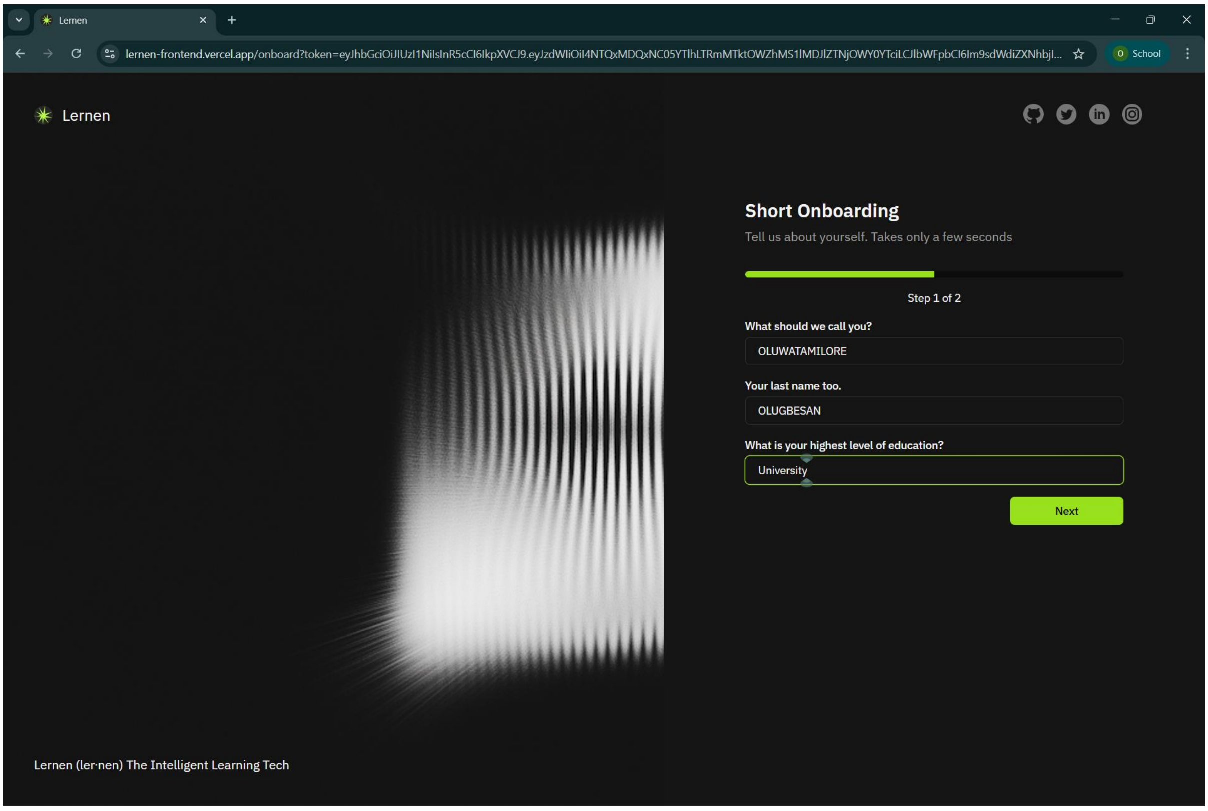

Figure 4.2 Onboarding page UI

4.3.2 Document Upload and Management

The document upload interface is integrated into the chat sidebar under a "Sources" section. Users can upload up to 5 PDF files per chat by clicking the "Add sources" button. Each uploaded document displays a checkbox for selection, the filename, a remove button, and an open button to view the document in the PDF viewer.

Upload progress is shown with a loading spinner and filename during processing. The system validates file types (PDF only) and file sizes (maximum 10MB) before uploading. Successfully uploaded documents are automatically selected for use in the RAG pipeline. Users can select or deselect documents using checkboxes to control which sources are searched when asking questions.

The sidebar also displays the user's recent chats in a "Recents" section, sorted by creation date. Each chat shows its title (derived from the first message or "New Chat" as default). Users can click on any chat to view its full conversation history. The sidebar includes a "New

Chat" button at the top to start a fresh conversation and a profile menu at the bottom for logout.

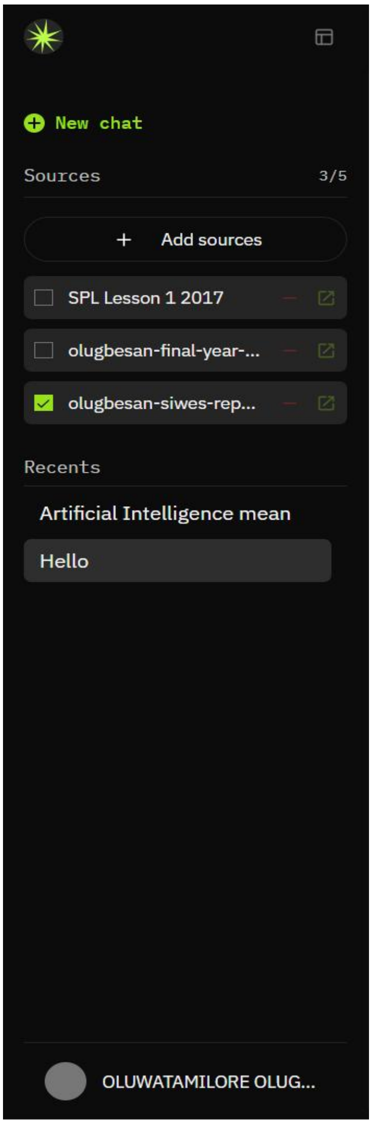

Figure 4.3 Sidebar UI

4.3.3 Question-Answering Interface

The question-answering interface consists of a message composer at the bottom, a scrollable chat area in the center, and an optional PDF viewer on the right. Users type questions in a message composer at the bottom of the page and submit by clicking the send button or pressing Enter.

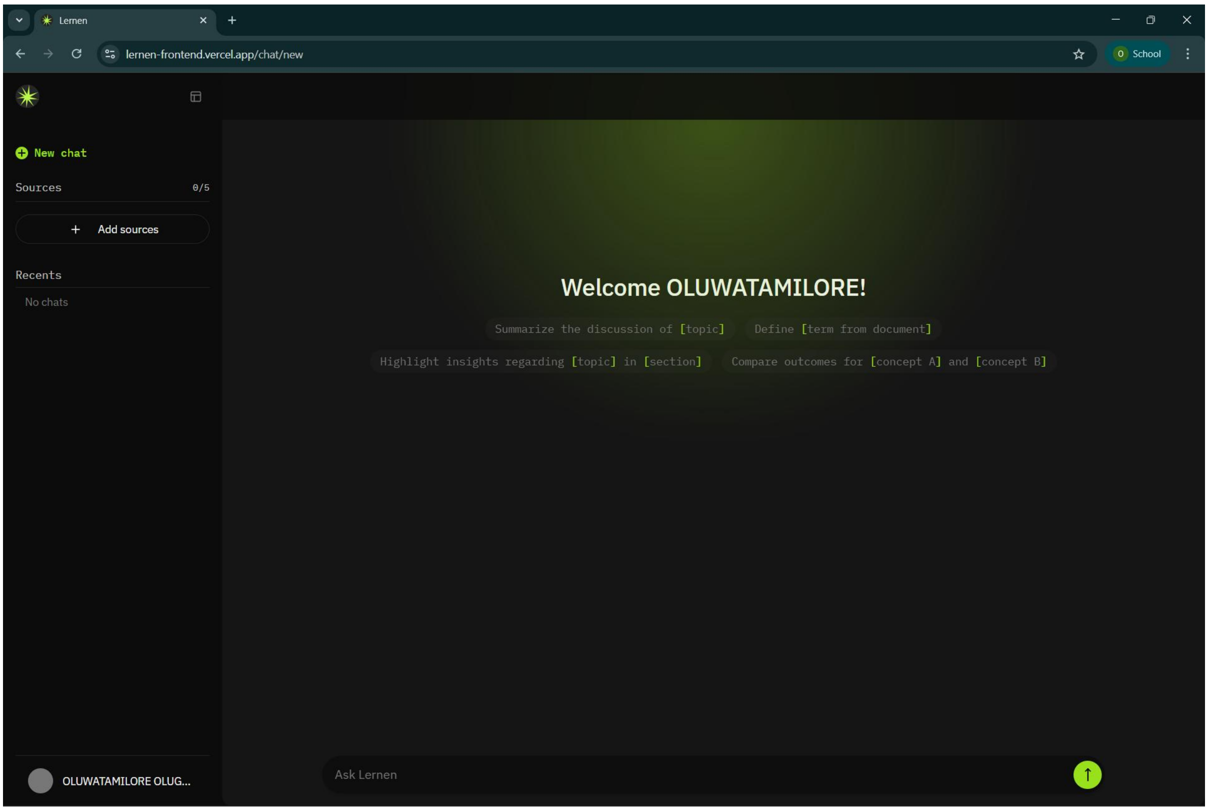

Figure 4.4 Chat Interface UI

The chat area displays conversation turns with user messages right-aligned and assistant messages left-aligned. Assistant messages include formatted markdown with support for code blocks, tables, lists, and inline formatting. Below each assistant message are feedback buttons (thumbs up/down) and a copy button to copy the message to the clipboard.

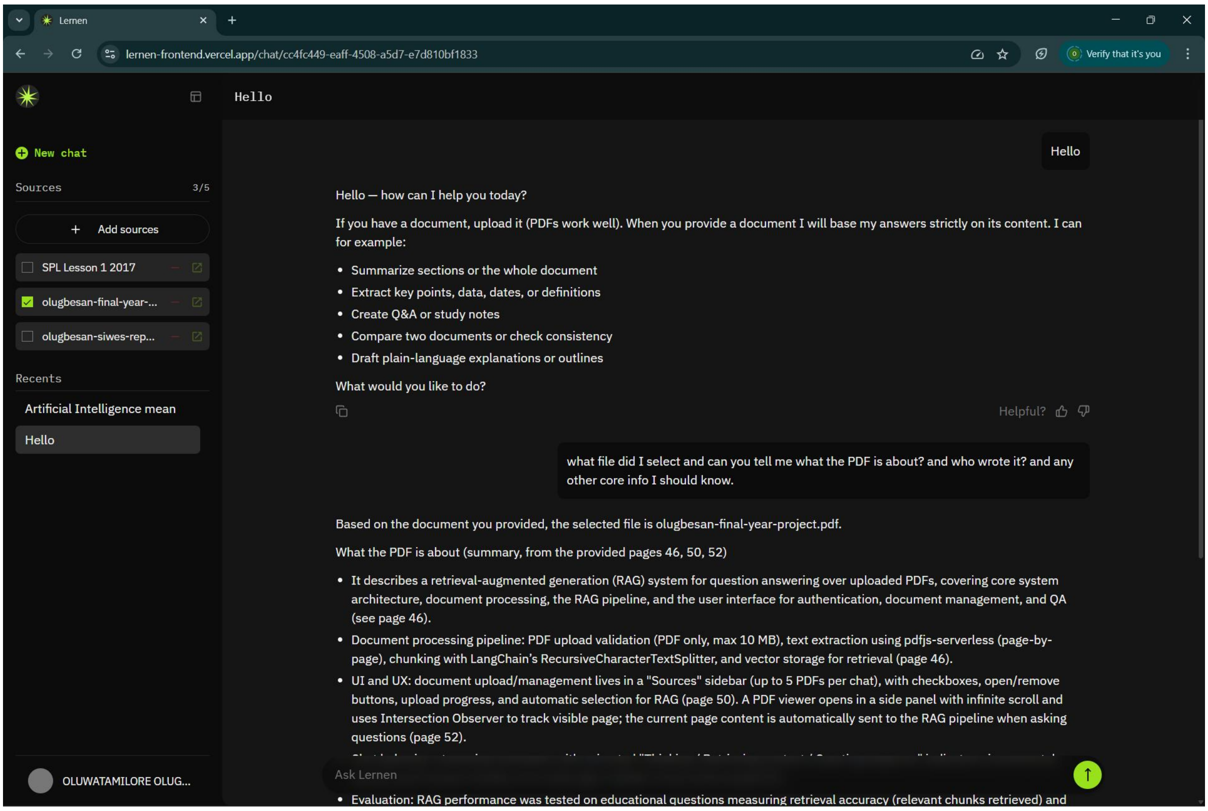

Figure 4.5 Chat interface with conversation turns UI

Streaming responses appear incrementally as chunks arrive from the server. If an error occurs, an error message is displayed with a retry button that allows users to resend their question.

The PDF viewer opens in a side panel when users click the open button next to a document. It displays the document page by page with infinite scroll, showing a page indicator at the top. The current page content is automatically sent to the RAG pipeline when asking questions, providing additional context for more relevant responses.

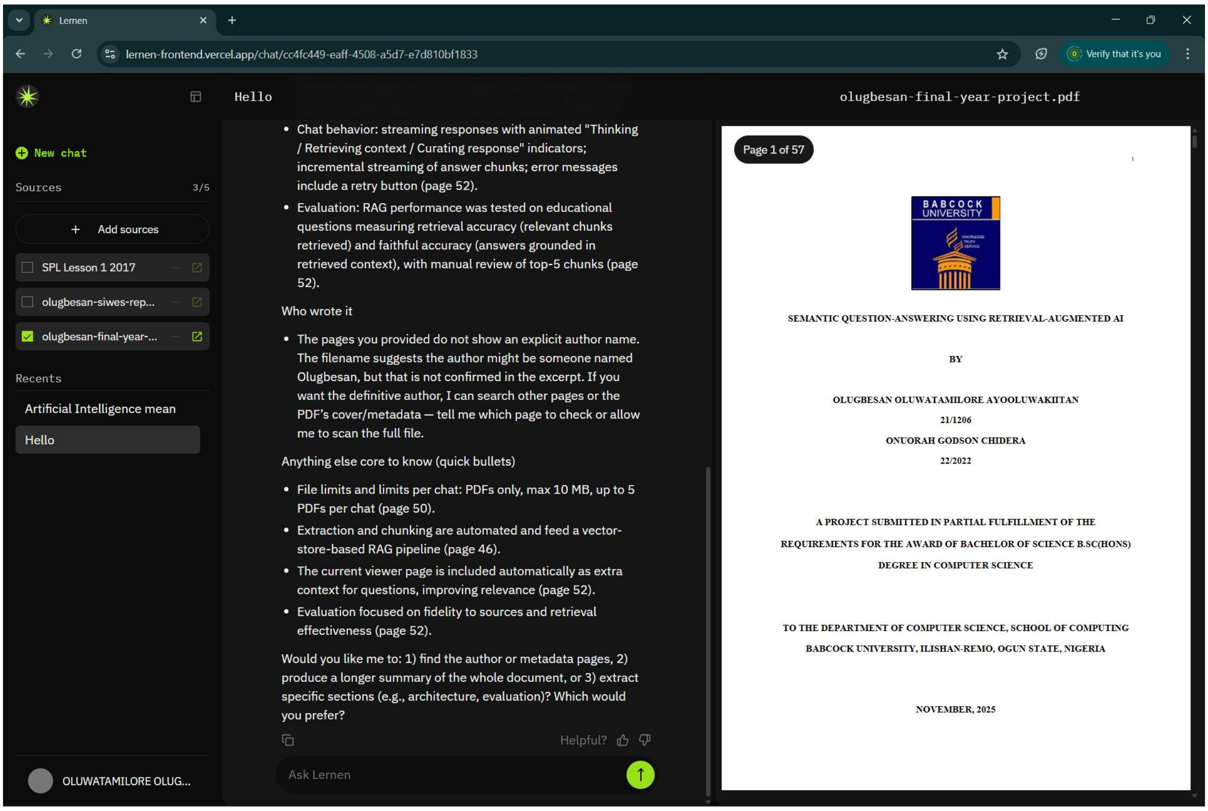

Figure 4.6 File viewer UI

4.4 Minimum System Requirements

The system is split into two deployable components: the frontend web application and the backend API server. Each has its own set of requirements.

Frontend (End User)

Users access the system through a web browser with no installation required. The minimum requirements are:

- A modern web browser: Google Chrome 90+, Firefox 88+, Microsoft Edge 90+, or Safari 14+
- JavaScript must be enabled
- A stable internet connection for API communication, document uploads, and response streaming

Backend (Deployment Environment)

The server environment requires:

- Node.js version 18 or higher
- A minimum of 512MB RAM, with more recommended under concurrent load

The following external services must also be provisioned and configured before deployment:

- A PostgreSQL database instance for persistent storage of users, chats, messages, and documents
- A Redis instance for caching stream sessions and managing short-lived request state
- A Cloudflare R2 bucket (or any S3-compatible object store) for PDF file storage
- A ChromaDB instance for vector embedding storage and similarity search
- A valid OpenAI API key with access to the embedding and GPT model families used by the system

4.5 Development Tools

The system was built across two separate codebases. The tools used in each are listed below.

Frontend Tools

- Next.js 15 with the App Router
- React
- TypeScript
- Tailwind CSS
- Radix UI
- react-pdf and pdfjs-dist

Backend Tools

- NestJS
- TypeScript
- Drizzle ORM
- PostgreSQL
- Nodemailer
- pdfjs-serverless
- OpenAI SDK
- ChromaDB client
- AWS SDK for S3
- Upstash Redis
- LangChain (textsplitters)
- @nestjs/throttler

CHAPTER FIVE: SUMMARY, CONCLUSION AND RECOMMENDATION

5.1 Summary

This study set out to design and implement a semantic question-answering system that addresses two well-documented limitations of large language models: the tendency to generate unverifiable responses, and the inability to reason over content outside a model's training data.

The resulting system, allows users to upload documents and query them conversationally. Rather than relying solely on a language model's stored knowledge, the system retrieves relevant content from the user's uploaded materials at query time and uses it to ground the model's response. This approach ensures that answers are traceable to specific source documents and pages.

Beyond the core retrieval pipeline, the study also produced a fully functional application with user authentication, document management, conversation history, streaming responses, and an integrated document viewer. Conversation continuity across long sessions was also addressed, ensuring the system remains coherent even as dialogue history grows.

5.2 Conclusion

The study demonstrates that retrieval-augmented generation is a practical approach to building educational question-answering systems that are both accurate and verifiable. By grounding responses in user-provided materials, Lernen produces answers that can be checked against the original source, directly addressing the reliability concerns that limit the use of general-purpose language models in academic settings.

The broader implementation also confirms that a dependable system of this kind requires considerably more than just the retrieval pipeline. Authentication, document storage, streaming infrastructure, and session management all represent non-trivial components that must work together for the system to be genuinely useful.

5.3 Recommendations

Based on the experience of building and testing this system, the following improvements are recommended for future development:

- **Multimodal document support:** The current pipeline is limited to text-extractable PDFs. Extending support to handle scanned documents, mathematical notation, figures, and formats such as DOCX and PPTX would meaningfully expand what the system can handle, particularly for science and engineering subjects.
- **Hybrid retrieval:** Combining vector-based search with keyword-based retrieval methods, and adding a re-ranking step, would improve results for domain-specific terminology that semantic search alone sometimes misses.
- **Adaptive learner modelling:** Introducing a component that tracks explored topics, identifies recurring misunderstandings, and adjusts the complexity of responses accordingly would move the system closer to a genuine tutoring tool rather than a retrieval assistant.
- **Reduced external dependency:** The system currently relies on third-party APIs for core functionality. Migrating to locally hosted open-source alternatives would reduce operational costs and make the system more sustainable at institutional scale.

5.4 Limitations of the Study

Several limitations were encountered during this research that are worth acknowledging.

The evaluation of the system was constrained by the absence of a formal user study. No assessment involving real students over a meaningful period was conducted. As a result, the system's actual impact on learning outcomes remains unmeasured.

Document support is also restricted to text-extractable PDFs. This is a notable constraint given that many academic materials, particularly older textbooks and journal articles, exist only in scanned form and would not be processed correctly by the current pipeline.

Finally, the system's dependence on third-party API services introduced practical constraints during testing. Usage limits restricted the volume of queries that could be evaluated, and the associated costs placed a ceiling on the scale of testing. These constraints would be more pronounced in a multi-user deployment.

5.5 Suggestions for Further Studies

This work opens several directions that future research could meaningfully pursue.
A comparative study evaluating the performance of open-source language and embedding models against proprietary alternatives, in the context of academic question-answering, would provide practical guidance for institutions weighing the cost and performance trade-offs of deploying such systems.

Research into retrieval pipelines capable of handling non-textual academic content, such as figures, tables, and mathematical expressions, would substantially extend the applicability of systems like Lernen, particularly in science and engineering disciplines where such content is central to the material.

A longitudinal study measuring actual learning outcomes among students using a retrieval-augmented QA system, compared against conventional study methods, would address what is currently the most significant gap in the literature. Most existing work evaluates these systems on accuracy benchmarks alone, without examining whether they genuinely support learning.

Finally, extending the architecture to support collaborative sessions, where multiple users contribute documents and query a shared knowledge base together, could open new directions for group-based and classroom-scale applications of this technology.

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